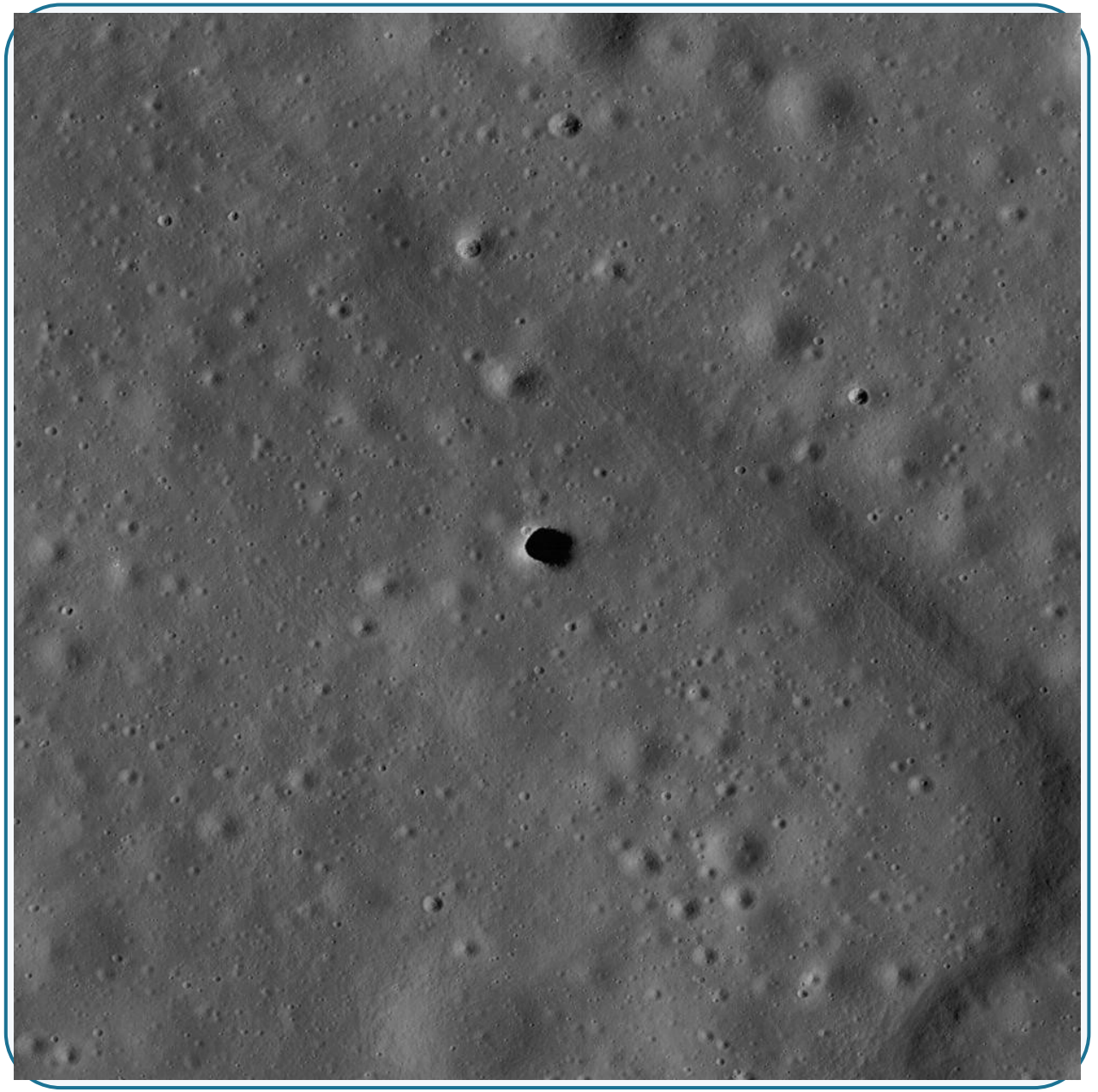
The frontier that becomes its own laboratory

AI in civilian space

Lecture 9

The solutions space forced us to invent — and that did not exist before it.



The collapse in launch cost — the lecture's foundation



An 18–20× drop — not gradual, but a step change: first-stage reusability

There is one named cause: first-stage reusability. 39 Falcon 9 boosters have flown an average of 14.9 times each — the rocket became an asset, not a consumable.

An important caveat for every number that follows. $2,700–3,140/kg is a price list divided by rated payload capacity, not the actual cost of a specific launch. For a satellite on a rideshare launch, the price is closer to $6,000/kg.

Harry W. Jones, «The Recent Large Reduction in Space Launch Cost», NASA NTRS

Satellites and data grew by an order of magnitude

STARLINK'S SHARE OF ALL ACTIVE SATELLITES

~65%

~11,100–11,150 satellites in one constellation

COPERNICUS (SENTINEL) ARCHIVE, 2018 → 2025

9.7 → ~91 petabytes

a ×9 increase in 7 years; EOSDIS by 2030 — ~600 petabytes



active satellites in orbit

A ~12× increase in 10 years — more tasks than operators able to handle them by hand

Algorithms didn't get smarter on their own — the number of tasks grew faster than the number of operators able to handle them by hand.

Union of Concerned Scientists · planet4589.org (J. McDowell) · CelesTrak · BryceTech · NASA EOSDIS · Copernicus/DLR

Who is launching, and how fast

COMPANIES WITH CONFIRMED FIRST-STAGE LANDING

4

SpaceX, Blue Origin — with reuse; LandSpace, Galactic Energy (China) — landing confirmed, reuse not yet

COUNTRY CONCENTRATION, SINCE START OF 2026

US 150 of 254 launches

China 76, Russia 13 — more than half the world's launches in one country

COMMERCIAL SPACE-INDUSTRY REVENUE, 2026

$544.3 billion

more than 80% of the entire global space economy (~$600 billion); ten years ago the ratio was reversed

RECORD SINGLE-BOOSTER TURNAROUND

9 days

landing → next launch (booster B1088); before 2017 this metric had no numerical value at all

10 years ago (2015), **not a single** company in the world had demonstrated a successful first-stage landing.

flightatlas.org (10.31.2026) · *OECD, «The Space Economy at a Glance 2026» (09.2026)* · *BryceTech*

The second round of the drop: Starship, plan and fact

STARSHIP TEST FLIGHTS AS OF 10.31.2026

14

the 14th reached orbit for the first time, deployed 26 Starlink satellites

BOOSTER CAUGHT / REUSED

3 times / 1 time

the second stage was never caught — there is no regular cycle yet

MUSK'S TARGET (PLAN, NOT FACT)

<$100 per kg

with full reusability + on-site propellant production — stated 07.29.2026

AEROSPACE CORPORATION ESTIMATE (REALISTIC RANGE)

$100–500 per kg

optimistic scenario — $67/kg after 9 cycles

An honest calendar caveat. Landing people on the Moon (Artemis, HLS based on Starship) has slipped to no earlier than September 2028 — a NASA Inspector General report (March 2026) says "years behind schedule".

The next attempt to catch both stages at once is no earlier than November 8, 2026, and that date may slip further.

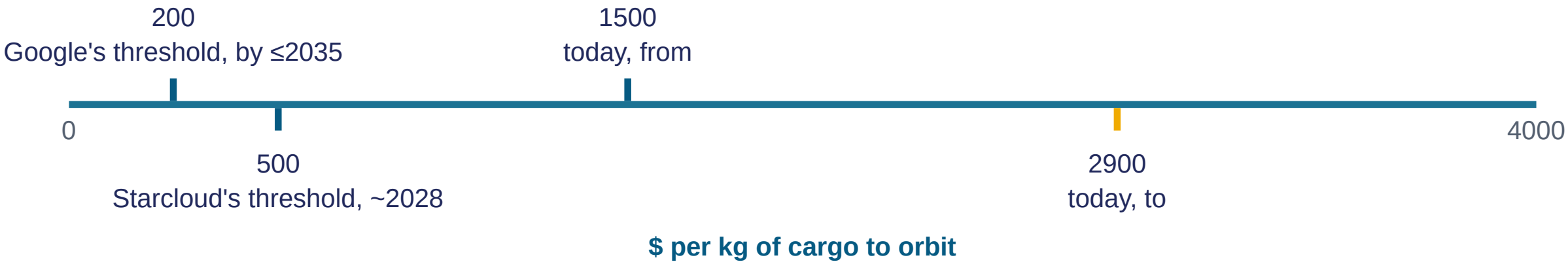
SpaceX, quarterly report 10.20.2026 · Aerospace Corporation / Ars Technica · Ark Invest · NASA OIG, report 03.2026

A threshold, not a promise

ORBITAL DATA CENTERS TODAY VS GROUND-BASED

2.5–3× more expensive

BCG: $660–750 million/MW against $230–300 million/MW on Earth



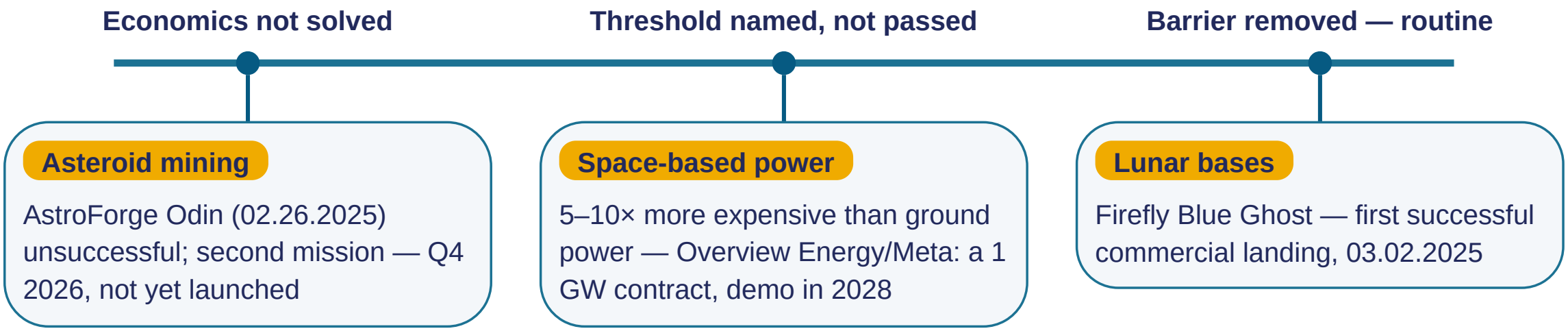
Today, several times above Google's threshold — the threshold is named, not invented

This is not "already worth it" — it's "will become worth it at such-and-such threshold". The difference between these two statements is what keeps the whole of Section 4 honest.

The reverse scenario. If cost freezes at the upper end of the range ($500/kg, not $67–100/kg), the drop from today's Falcon 9 will be not 30× but 5–6×, and Google's threshold will move years further out than 2035.

Google (Project Suncatcher) · *Starcloud* · *BCG, report 08.2026*

Why right now — three frontiers



track maturity, 2025–2026

All three share the same starting point — a 1970s idea; the shared trigger is the collapse in launch cost. That alone doesn't solve the track's economics

A shared pattern: the drop in launch cost **does not solve the economics** of each track on its own — it lowers the threshold for even attempting it.

Overview Energy/Meta (04.27.2026) · *Firefly Aerospace, Blue Ghost Mission 1 (03.02.2025)* · *AstroForge*

What exactly space pays with for the pressure of being a frontier

What the frontier pays with				
	Distance 3–22 min to Mars	Data volume ×12 satellites, ×9 archive	Human time hours of CFD, routing, forecasting	New capability computation in orbit

Four places where being a frontier demands payment — in distance, data, human time, a new task

Space is a frontier: extreme conditions make off-the-shelf solutions insufficient and force the creation of new ones — invented precisely under this pressure, not borrowed from softer conditions.

The familiar, easier answer: engineering solutions come from the lab, space just applies them. The next slide tests that answer with one concrete case.

From an earthly mine to a lunar lava tube

DARPA Subterranean Challenge, 2018–2021

Mine

a competition for autonomous reconnaissance of mines and collapsed buildings — GPS and comms don't work underground, same as in space

LAMP 2.0 architecture

LAMP 2.0

8+ robots stitch maps together over kilometer-scale distances — measured at the earthly competition

COROB-X — lunar lava tube

Moon

carried over in spirit, not in code; tested on Earth, Lanzarote, 02.2023 — never flown to the Moon

transfer stage of the solution

One architecture, three stages: from a mine competition to a lunar demonstrator

The transfer: the lecturer's own testimony.

What follows is a case from the lecturer's own practice about carrying a solution from one frontier to another; told orally, with no text on screen.

Closing line. The familiar answer: "solutions come from the lab, space applies them". The lecture's question: what happens when the frontier itself becomes the laboratory?

DARPA Subterranean Challenge (2018–2021) · *JPL/Caltech/MIT/KAIST/Luleå, the NeBula/CoSTAR family*

The lecture's question

Which decisions did space force us to invent —
and did they exist before it demanded them?

The answer — at the end of the lecture, named case by case for each of the four sections.

The map of the evening: four fronts

Section 1 — maximum autonomy: no human in the loop at all. Section 2 — minimum autonomy: a human controls, just at a distance. Section 3 — not about autonomy at all: a model designs and forecasts. Section 4 — a mirror of the whole lecture: it's not space getting smarter, it's AI moving into space.

Next — Wrap-up (registry, answer to the question, the tally) and Closing (the debt carried forward).

No.	Section	Term	What it hinges on
1	Autonomous robots in space	autonomous robotics	the machine decides and acts on the spot with no dispatcher
2	Remote control	teleoperation	a human controls at a distance — where that stops working
3	Design and new capabilities	AI-assisted design and forecasting	a model does work that simply didn't exist before
4	Space for AI	orbital computing	computation moves to where there's more energy

How to read this lecture's numbers

A separate rule — the denominator. "93% hit rate" without stating "of what" means nothing. Where the denominator can't be found in open sources, the lecture names that as a gap rather than inventing a figure.

Type of number	Example	How it's presented in the lecture
Fact-number	24%→93% AEGIS; Dragon routine since 2020	always with a source and a date
Threshold-number	$200/kg (Google's threshold); <$100/kg (Starship's target)	explicitly a plan/goal/condition, not achieved
Analyst's-estimate-number	$100–500/kg Aerospace; $115–200 Ark Invest	always a range with a source, never a single figure

The breakdown's form: five steps

Steps 1–3

Cost of both errors → Both classical modes (a/b/c) → The model vs the modes

Steps 4–5

How it works: input/output/name → What moved over + what it gave/cost

The form of the breakdown — applied identically to all 13 cases in the lecture

Step 2's column — applied identically throughout the lecture: (a) both modes are given a number in the same unit · **(b)** the second mode is named but has no number of its own · **(c)** a second mode simply doesn't exist for this task, and the reason is stated.

A gain measured from zero (ignoring what the classical mode could already do) is not accepted.

1. Autonomous robots in space

The machine decides and acts on the spot — with no dispatcher at every step.

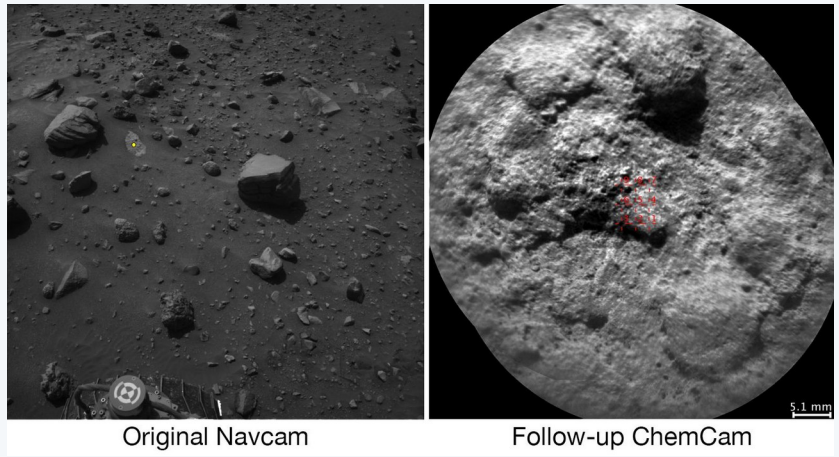
In order:

AEGIS — autonomous selection of scientific targets on Mars

SPOC — deep learning for labeling Martian soil

COROB-X — three robots in a lava tube

LeoLabs — radar prioritization and collision warnings



Original Navcam | Follow-up ChemCam

The stage: autonomous robotics

TERM

autonomous robotics

deciding and acting with no dispatcher at every step of the cycle

DATA

environment + own state

soil, terrain / position, charge, remaining range

DECISION

a specific physical action

dig here, drive this route, photograph this object

WITHOUT AUTOMATION

a human on the spot

an astronaut in a suit, an engineer on a tether — limited by risk and the length of the excursion

This section comes first, and not arbitrarily. This is the lecture's point of maximum autonomy, against which the rest of the sections can later be compared.

Three ways of getting data — all onboard the machine itself: cameras (stereo pairs), LIDAR, inertial sensors.

Where the classical approach handles it, where a model is needed

The classical approach handles it, no training

- Terrain traversability, cave mapping (AutoNav/GESTALT)
- Falcon 9 landing — convex optimization (G-FOLD)
- The physics of soil and optics is known in advance

Needs a trained model

- Recognizing an artifact against Martian dust
- Features aren't known in advance, statistics from examples
- A lightweight classifier inside an otherwise classical pipeline

The strongest example of this distinction lies outside surface navigation entirely. Falcon 9's first-stage landing (G-FOLD) is convex optimization, **not a neural network**. It visually looks like a miracle of autonomy — and it is one, but its autonomy is built from math, not from learning.

How the methods work: geometry and random forest

STEREO VISION (PARALLAX)

disparity → range by geometry

two cameras on a known baseline, similar triangles, no trained parameters

RANDOM FOREST

dozens to hundreds of decision trees

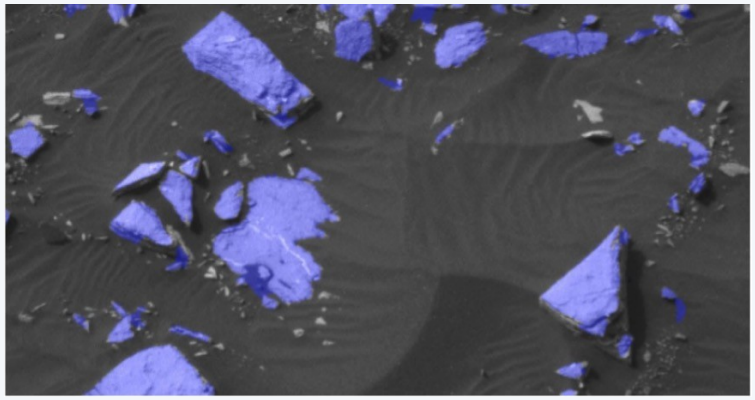
bagging + a random subset of features at each node

THE KEY PROPERTY OF A RANDOM FOREST

interpretability

a tree can be read all the way through, step by step — not a black box

Why exactly this mechanics matters: it's what holds up AEGIS. Distance to the target (Block 1) is one of the features by which interest is ranked (Block 2). Both blocks are below, before we move on to the case study itself.



How the methods work: a map without GPS and a single-glance detector

CNN DETECTION (THE YOLO FAMILY)

coordinates + class in a single pass

'you only look once' — the whole frame's grid at once, not window by window

input

Local odometry

estimating one's own displacement from sensors

without it, the robot has no map of its own

work

Match search between robots

finds places where different robots were at the same spot

even if the robots never met directly

output · bottleneck

Joint pose-graph optimization

matches become edges of a shared graph

minimizes the total inconsistency across all edges

Three tiers, not two — each one closes a gap the previous one leaves.

A detector ≠ geometry. The detector decides "what this is and where on the frame" (pixels); geometric localization in space is, again, the classical approach, not a neural network.

Example: AEGIS — autonomous selection of scientific targets on Mars

COST OF A MISS

1 lost sol

re-choosing a target costs another full communication cycle with Earth

CLASSICAL MODE (BEFORE AEGIS)

24% hit rate

targeting by Earth command based on the previous image

THE MODEL (AEGIS, ROCKSTER + RANDOM FOREST)

93% hit rate

routine on Curiosity since 2016 (first on Opportunity, 2010)

HOW IT WORKS

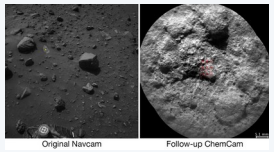
NavCam stereo-pair input → Rockster → features → random forest → ChemCam

range — by geometry (Block 1), ranking — by random forest (Block 2)

What moved over. Productivity grew almost fourfold — a scientist sets the target profile in advance, once, instead of choosing every target personally.

An honest gap. Sources don't disclose the feedback loop or the composition of the random forest's training set.

Image classification.

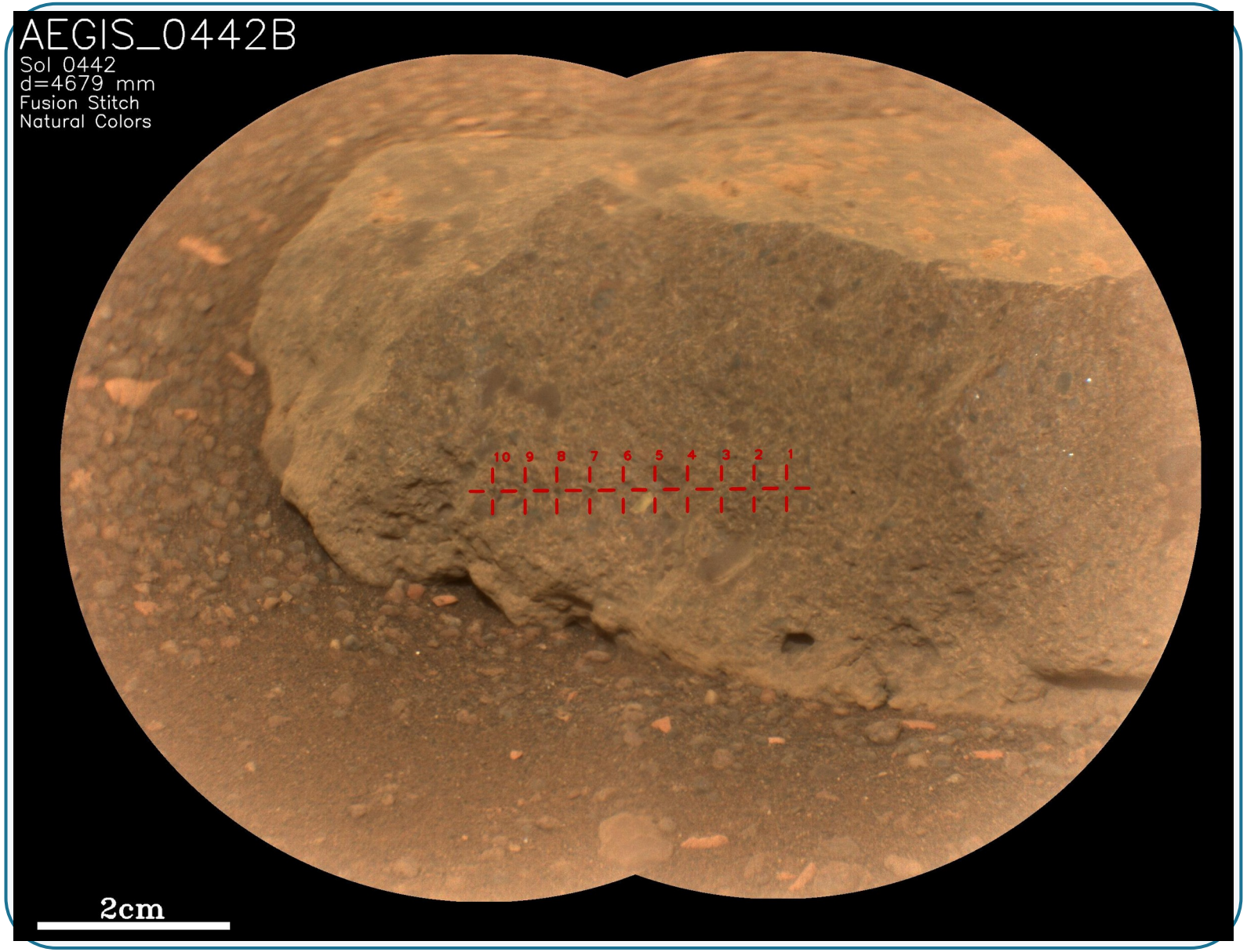


Francis et al., Science Robotics 2017 · *JPL, AEGIS project page*

AEGIS on Perseverance

Not a one-off trick on a single rover. AEGIS first fired on the Perseverance rover's SuperCam instrument in May 2022 — the same algorithm, a different rover, a different instrument.

It's the architecture that transfers, not the number. Rockster and the random forest stay the same; the target profile is tuned for the new instrument from scratch.



NASA Science, PIA25289, 05.31.2022 · Photo: NASA/JPL-Caltech/LANL/CNES/IRAP, first use of AEGIS on the Perseverance rover's SuperCam, 05.18.2022

How AEGIS is built into the rover

input

Input: NavCam stereo pair

range to the object — by geometry, via parallax, not a neural network

Rockster searches for candidates by brightness contrast and edges

work

Work: random forest ranks them

the classifier scores candidates against the target-of-interest profile

the scientist sets the profile in advance, on Earth, once

output

Output: coordinates for ChemCam / SuperCam

the shot happens onboard, with no round-trip to Earth for approval

the choice of the specific target in the moment belongs to the algorithm, not a human

Where a human used to stand. Before AEGIS, a scientist chose every target personally, from the previous cycle's image — that is, in the spot held by this diagram's third block.

What the automation did. The third block moved onboard: the target's coordinates go straight to ChemCam or SuperCam, with no waiting for a command from Earth.

Example: SPOC — labeling Martian soil with a neural network

COST OF ERROR

a stuck rover

loose sand mistaken for firm soil — risk of months of recovery or losing the vehicle

CLASSICAL MODE

the driver's visual judgment

named, but no precise figure exists in open sources

THE MODEL (SPOC-HIRISE / SPOC-NAVCAM)

98% accuracy

a deep CNN right on pixels, already used by Curiosity and Perseverance drivers

HOW IT WORKS

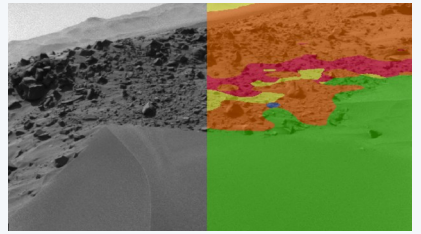
HiRISE/NavCam input → patches → CNN → soil-class map

a ground tool today; forward-looking — SPOC-Lite (CPU) and SPOC-HPSC/SNPE (GPU) for a future onboard version

The feedback loop is explicitly closed — unlike AEGIS and COROB-X: the AI4Mars project (Zooniverse) is recruiting volunteers to label Perseverance images right now.

A forward-looking part. SPOC-Lite and SPOC-HPSC/SNPE are for future missions, **not deployed** on flying rovers.

Image classification.



[NASA JPL, SPOC project page (nasa-jpl.github.io/SPOC)](nasa-jpl.github.io/SPOC) · [AI4Mars/Zooniverse]

Soil for SPOC

Four surface classes, not an abstraction. Soil, bedrock, sand, and large rock — exactly the types AI4Mars asks volunteers to label.

This data is what holds up the 98% accuracy. Without a steady stream of newly labeled data, SPOC can't learn new Martian terrain.



NASA Science, PIA23897, 06.12.2020 · Photo: NASA/JPL-Caltech, Martian soil types from the AI4Mars project's data

How SPOC is built into route planning

input

Input: rover camera images

orbital HiRISE and onboard NavCam — two sources of frames

the image is sliced into patches for pixel-level classification

work

Work: CNN soil classification

a deep convolutional network determines surface type right from pixels

no manual feature extraction — unlike AEGIS's random forest

output · with the driver

Output: a traversability map

the map feeds into route planning, it doesn't replace the decision

whether to drive a section or not is decided by the rover driver on Earth

The model labels, it doesn't decide. The soil-class map is a recommendation for planning, not a command that goes straight to the hardware.

The final authority is named directly. The decision to drive or not drive a specific section stays with the rover driver on Earth — same as before SPOC.

Example: COROB-X — three robots in a lava tube

COST OF RISK

a ceiling collapse with no warning

a human on a safety tether risks their life; three robots do not

CLASSICAL MODE

does not exist

no industry on Earth had solved this task before this demonstrator

AUTOMATION (SHERPATT + COYOTE III + LUVMI-X)

4 phases: mapping → sensor cube → tethered descent → 3D reconnaissance

a three-tier SLAM in the spirit of the DARPA mine architecture + a YOLOv4 CNN layer

HOW IT WORKS

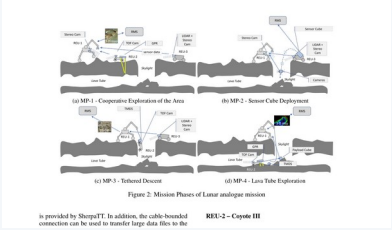
visible+thermal spectrum → YOLOv4 → EaRLaP → geometry

the arbitration is the EaRLaP pipeline, filtering out false detections

What moved over. Safety unambiguously improved — three robots don't put a human life at risk. Complexity grew just as unambiguously: one human with a flashlight → three machines, a tether, a surface base station.

An honest gap. Sources don't give a numerical measure of map quality. The system has never flown to the Moon or Mars.

Image recognition.



DFKI (Germany) together with Space Applications Services · field testing, Lanzarote, 02.2023

COROB-X's field test

A field test site, not the Moon. Lanzarote, Canary Islands — a lava tube chosen specifically for its geological resemblance to lunar skylights.

In the photo: the Coyote III rover, not a human. The tether holds the machine, not a climber: the same descent that only a roped human used to make.



DFKI, press release 03.08.2023 · Photo: DFKI, Tom Becker

How COROB-X's three robots build one map with no GPS

input

Input: three robots, three different maps

SherpaTT and LUVMI-X are outside, Coyote III is inside the tube

each computes its own displacement from sensors — with no GPS

work

Work: matches between the robots' maps

finds places where different robots were at the same point

works even if the robots never met directly

output · bottleneck

Output: one map of the tube, with no connection to the surface

matches become edges of a shared pose graph

the three maps stitch into one during the descent itself

The mechanism is on the neighboring slide, this one is the application. The three SLAM tiers are already broken down on the methods slide; this slide shows exactly how they're built into the COROB-X mission.

Three robots of different designs, one map. SherpaTT and LUVMI-X work outside, Coyote III inside the tube; each with no GPS and no direct link to the others during the descent.

Example: LeoLabs — radar prioritization and collision warnings

COST OF A MISS

loss of a satellite + a debris cascade

Kessler syndrome: threatens every neighboring satellite at the same altitude

COST OF A FALSE ALARM

fuel depletion + noise for the operator

unnecessary maneuvers, after which a real warning gets lost

CLASSICAL MODE

government networks, named, no figure found

less frequent radar imaging, a coarser detection threshold

THE MODEL (LEOLABS)

up to 400% more warnings

11 radars at 7 sites, detection from 2 cm; ~$60 million in annual contracts in 2026

An honest caveat. Sources don't state how many maneuvers satellites actually performed in response to these warnings in 2026 — a warning issued ≠ a maneuver performed.

For contrast. With Starlink, the automation both warns and maneuvers itself — there's no third party at all.

Classification (prioritization) on structured data.

LeoLabs (eoportal.org, leolabs.space/conjunction-alerts, SpaceNews)

How LeoLabs is built into the warning system

input

Input: a growing catalog of objects

11 radars at 7 sites, detection of objects from 2 cm

the orbital-debris catalog grows with every pass

work

Work: a prioritization model

ranks which object is most worth the radar observing next

predicts the probability of getting a good-quality measurement

output · an honest boundary

Output: a close-approach warning

up to 400% more warnings than the previous mode

whether to maneuver or not is decided by the satellite's operator, not LeoLabs

The model doesn't predict a collision directly. It solves a different task — where to point the radar next, given limited observation time.

The boundary is stated directly, with no guesswork. LeoLabs issues the warning; the decision to maneuver belongs to the specific satellite's operator, not to LeoLabs.

Section 1 mentions

- **AutoNav/GESTALT** — the classical approach, not a neural network: drives Perseverance for **90%** of its route (against 6% for Curiosity)
- **NASA IPEx** — an autonomous lunar excavator, **with not a single trained model**: up to 10,000 kg of regolith per lunar day, TRL5 (technology readiness level 5 of 9 — tested under real conditions, not just in the lab)
- **ICON Project Olympus** — laser sintering of regolith; the claimed AI quality-control layer is **not publicly disclosed**
- **Starlink** — ~300,000 autonomous maneuvers a year **on its own, with no third party**: it warns and executes within one organization
- **JPL GIMLI / Moon Diver** — proposed missions into lava tubes, early selection stage, **not flying vehicles**
- **ISRU autonomy** — oxygen extraction TRL6, water extraction (LADI) TRL4

NASA JPL · NASA NTRS (IPEx) · ICON · Starlink

What it gave and what it required — Section 1

Gave

- Scientific productivity ×4 (AEGIS)
- Soil labeling at 98% accuracy (SPOC)
- Reconnaissance of unreachable places (COROB-X)
- Up to 400% more warnings (LeoLabs)

Required

- Pre-certified thresholds (geometry)
- More complexity: 3 robots instead of 1 human
- $60 million/year of radar infrastructure
- The decision to respond stays with the operator

Grid cell	Case study
Classification × image	AEGIS, SPOC (at different depths)
Recognition × image	COROB-X
Classification × structured data	LeoLabs

In sum: the gain is measurable and large where the task is already solved and confirmed by working operation (AEGIS since 2016, SPOC as a ground tool already today, LeoLabs as a commercial service). Where the technology hasn't been tested on the real Moon or Mars, that's stated directly.

Section 1 conclusion

AEGIS

productivity grew almost 4×

range geometry + a lightweight classifier on top of it, routine since 2016

SPOC

98% soil-labeling accuracy

a deep network on pixels — currently a support tool on the ground, not an onboard module

COROB-X

cave reconnaissance with no human inside

three robots and a tether instead of one human with a flashlight — a task with no classical equivalent

LEOLABS

up to 400% more close-approach warnings

at detection of objects from 2 cm — the satellite operator still decides whether to respond

Geometry is solved with no training — by classical calculation. Recognizing an object against noise is solved by a lightweight model inside a classical pipeline, not replacing it. A wholly new task — cave reconnaissance with no GPS or communication — has no classical equivalent at all.

Next — Section 2: the same task, a different degree of autonomy.

2. Teleoperation

A human is in control — but from a distance. Where does this stop working?

In order:

Dragon — autonomous docking to the ISS

Predictive latency compensation — a trained model instead of a classical filter

Astrobee — ML-assisted piloting



Scene: the crew watches the ship park itself

Four astronauts in the Dragon capsule, 200 meters from the ISS. The commander keeps his hands on the screen — not to fly the ship, but to be ready to take control.

The ship docks itself to the docking port without a single command from the screen: rangefinders measure distance dozens of times per second, and automation calculates the thrust. A few minutes later — the soft click of the latches.



A routine operation since 2020 — not a test, not a demo.

SpaceX/NASA, Demo-2 mission (2020)

Stage: teleoperation

TERM

teleoperation

controlling a system from a distance, the decision stays with the human

DATA

real-time video feed + telemetry

the question: how long a human can stay in the loop at a given delay

DECISION

confirm or cancel an action

as far as the communication delay allows

WITHOUT TELEOPERATION

a human physically on site

safer in terms of control, requires presence and risk

This section follows Section 1 for a reason: it shows where the boundary of autonomy has not yet been crossed — a human holds it deliberately, not by default.

Where a human is needed, where the boundary has already shifted

TO THE ISS

a fraction of a second

the same order as the Starlink delay in telesurgery — covered later in this section (~130 ms)

TO THE MOON

1.3 seconds

plus the controller's own operational delays

TO MARS

3–22 minutes

depending on the relative position of the orbits — 2–3 orders of magnitude more than the lunar delay

This is a physical, not an organizational, reason why Section 1 and Section 2 need to be two separate sections at all — a property of the geometry of distance that no communication channel can outrun faster than the speed of light.

Where an event unfolds faster than a human can react — control is handed to automation **in advance**, with the right to intervene, not after the fact of an accident.

How the methods are built: the classical controller and the warm start

base

Controller (feedback)

measures → compares → computes a command, dozens of times a second

stability guarantees — strictly within the given conditions

acceleration

Warm start

the model suggests a starting point for the trajectory search

the final guarantee does not change — the controller checks the solution

separate mechanism

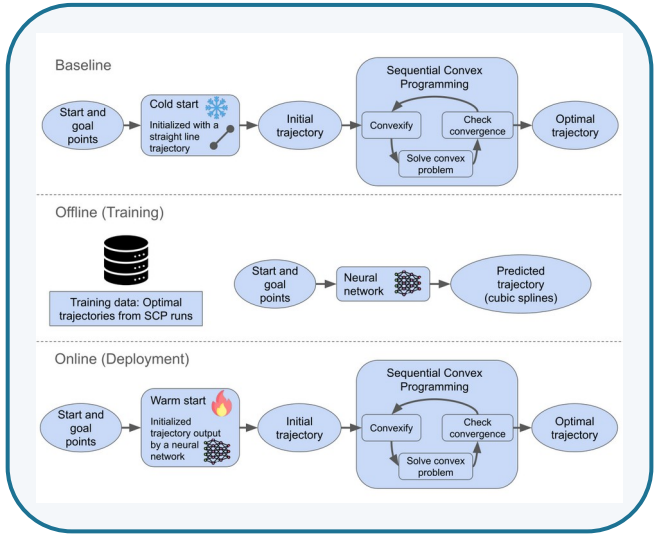
Predictive latency compensation

the model predicts the scene across the interval of the communication delay

changes what the operator sees, not what they decide

The controller's key property. Provable mathematical stability guarantees — but strictly within the conditions it was designed for in advance.

A warm start in the worst case — does not make the final decision any less safe, only spends more iterations refining it.



How predictive latency compensation is built

input

Input: delayed stream

video and recent history of the scene's motion from the remote site

the frame is already stale by the amount of the communication delay

work

LSTM or temporal convolutional network

the LSTM steps through frames one at a time and keeps a memory

the TCN looks at a window of past frames at once, without memory

trained on recorded past sessions of scene motion

output · honest gap

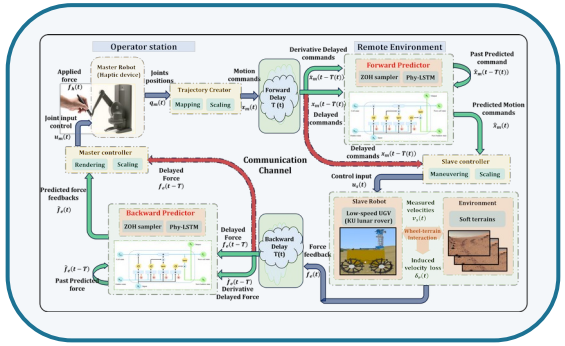
Output: frame for the operator

the predicted, corrected frame replaces the raw video on the screen

the sources name no formal threshold for checking the prediction

Changes what the operator sees, not what they decide. Neither the controller nor the warm start is involved here — all movement commands stay with the human.

An honest gap. The sources name no formal threshold for checking the prediction before it is shown.



Example: Dragon's autonomous docking to the ISS

COST OF MANUAL ERROR

millisecond precision at the edge of motor control

speed, angle, and position must coincide with hard-to-achieve precision

COST OF FULL AUTOMATION WITH NO MANUAL INTERVENTION

an unforeseen scenario

any system designed for a finite set of scenarios risks running into a new one

CLASSICAL MODE

manual docking, Shuttle / Soyuz

named, but the hours of preparation and attempt statistics are not published

AUTOMATION (NASA DOCKING SYSTEM)

autonomous from 220 meters

routine since 2020; manual intervention available at any moment

220 meters is not a physiological limit of reaction time, but an engineering decision: where automation is reliably more precise than a human.

What moved over. Precision of the final approach — to automation; responsibility — stays with the human: he monitors and can intervene, though he almost never does.

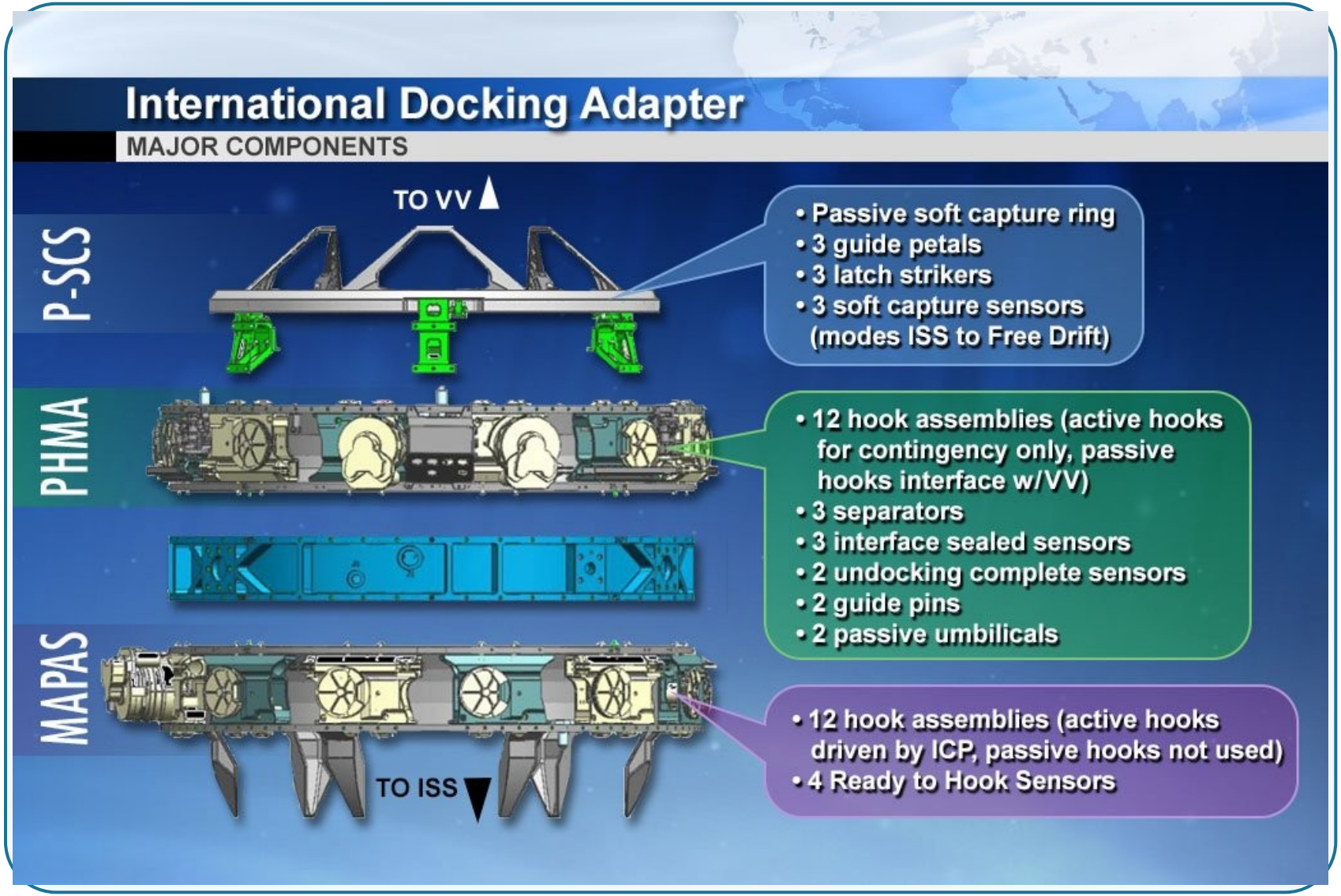


NASA/SpaceX, Demo-2 mission (2020)

Dragon's docking port

Real hardware, not a diagram. The ring, alignment sensors, soft capture latches — this is exactly what physically receives Dragon on the final meters of approach.

This is exactly where the precision named on the neighboring slide arrives. Millimeter tolerance in angle and position is not an abstraction, but a requirement of this specific mechanism.



Wikimedia Commons, NASA, 08/19/2016 · Photo: NASA, main components of the docking mechanism, public domain

How Dragon's docking automation is built

input

Input: laser rangefinders and video

per public data — LIDAR and cameras on approach to the station

range and approach angle are measured in real time

work · honest gap

Work: internal control logic

the NASA Docking System's architecture is not publicly disclosed by SpaceX

trained statistically or computed classically, like G-FOLD — unknown

output

Output: autonomous docking from 220 meters

routine since 2020, precision at the edge of human motor control

the crew can take manual control at any moment

An honest gap — not a research shortfall, but a fact about Dragon. SpaceX does not publish what happens inside the middle block of this diagram, and the chapter states this directly instead of guessing.

The outer blocks are known precisely. Input — sensors per public data; output — routine autonomous docking since 2020, with manual intervention standing ready.

Example: predictive latency compensation

COST OF OVER-CAUTION

slow, often-stalling work

the operator waits to make sure the picture has 'caught up' with reality

COST OF RISK

acting on a stale picture

correcting an already-made move in jerky steps

CLASSICAL MODE

extrapolation at constant speed

works while motion is uniform; errs on sharp changes

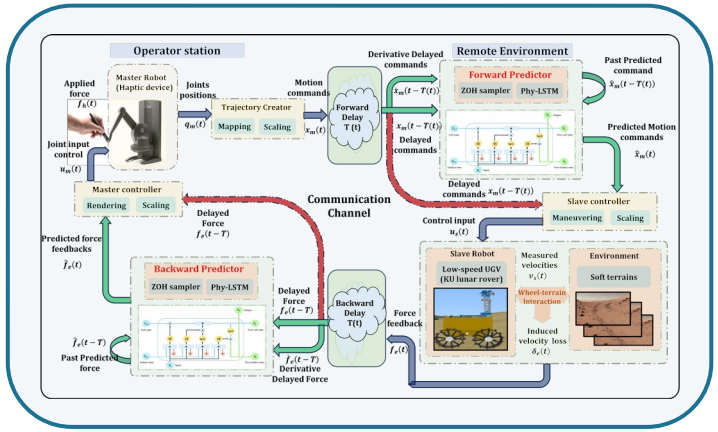
MODEL (LSTM / TCN)

reduces task time by up to 26%

lunar analog facilities — a research stage, not routine practice

Honest status: a research stage, not deployed practice. Tests are at lunar analog facilities, not on a real mission; the lunar analog is confirmed by a single journal source, not a series of independent repetitions.

Prediction on video — the first appearance of the "video" modality in the lecture's grid.



Frontiers in Space Technologies 10.3389/frspt.2026.1920649 · *ScienceDirect S1526149226000639*

The LUNA analog facility, Earth

Honestly: this is Earth, not the Moon. The LUNA facility (Germany, ESA/DLR) is covered with lunar regolith simulant — a test at a site resembling the target, not the target itself.

This is exactly where what was broken down nearby is tested. Predictive latency compensation — a research stage, not deployed practice on a real mission.



Wikimedia Commons, Raimond Spekking, 09/25/2024, CC BY-SA 4.0 · Photo: Raimond Spekking, Wikimedia Commons, CC BY-SA 4.0 — a demonstration by astronauts at the LUNA facility (Earth)

Example: Astrobee — ML-assisted piloting

COST WITHOUT AN ASSISTANT

delay and load on the operator

multiple small course corrections from Earth

COST OF FULL AUTONOMY WITHOUT A CHECK

collision risk

the station's cramped interior space

CLASSICAL MODE

manual remote piloting

named, the number of interventions per maneuver is not published

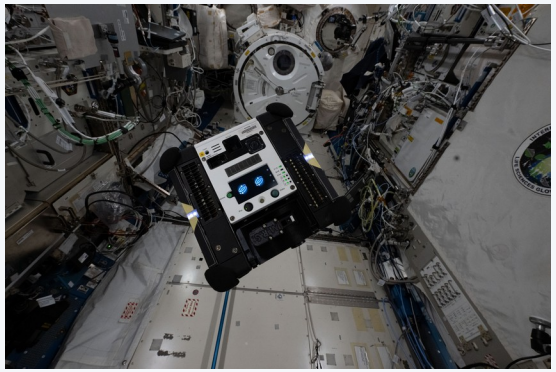
MODEL (WARM START)

TRL5, tested aboard the ISS

Stanford + NASA, 12/2025; the robot was handed to Arkisys in summer 2025

Arbitration is not human, but machine. The classical controller automatically refines or rejects the model's proposal on every cycle, before it becomes a thrust command.

The only breakdown in the section where the sources name the developing organization (Stanford) separately from the operating organization (NASA).

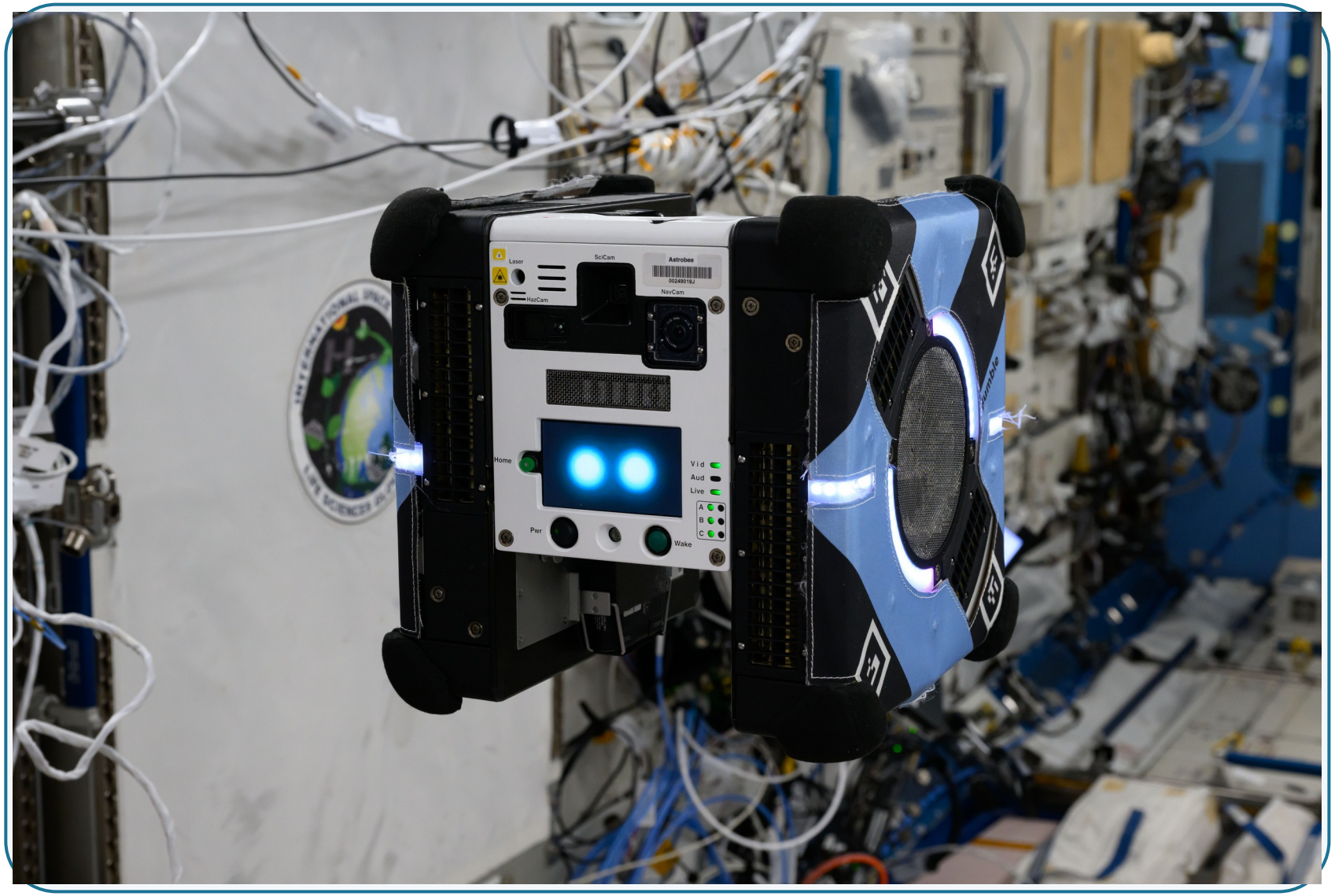


Stanford + NASA, demonstration 12/2025

Astrobee on the ISS

A cube-shaped body with no tether and no legs. Astrobee holds itself in the air with fans that push air out through side nozzles — not magnets, not wheels, not grabbing the walls.

This exact free flight is what the warm start refines. The model proposes a flight path through the station's module, and the classical controller checks it before every maneuver.



NASA Johnson Space Center, Wikimedia Commons, 05/01/2024 · Photo: NASA Johnson Space Center, public domain

How the warm start is built into Astrobee

input

Input: free flight inside the station's module

current position and target point, fans instead of a tether

the space is cramped — risk of collision on a maneuver

work

Work: the model proposes, the controller refines

the model, on statistics of past trajectories, suggests a start

the feedback controller refines it before the maneuver

output · bottleneck

Output: thrust command to the fans

the controller carries the final safety guarantee

an unlucky start only costs more iterations, no worse off

The warm start's design is on the neighboring slide; here — what it is built into. This mechanism suggests the starting point of the trajectory specifically for Astrobee's free flight inside the station's cramped module.

The final decision is not the model's. The controller checks and refines the proposed trajectory before every command to the fans.

Mentions from Section 2

- **Exosuit with "artificial muscles"** (Bristol) — tested only in Earth analog missions, **never flew** in space
- **ESA exosuit** — force feedback: the operator feels the tool's resistance physically, not just visually
- **Earth–ISS delay** — a fraction of a second, the lower bound of the range that Section 1 stretches out to 22 minutes
- **Telesurgery over Starlink** — delay ~130 ms; lung resection on an animal model, performed remotely; **there is no trained model in the decision loop at all**

University of Bristol · ESA · PMC (telesurgery)

What it gave and what it required — Section 2

Gave: access to operations without physical presence at the event — the surgeon stays on Earth, the operator at the console, the crew a witness at the moment of docking.

Required: being willing to accept that not everything is routine practice (latency compensation is only an analog); deciding in advance exactly when automation acts without confirmation.

Grid cell	Breakdown
Forecasting × video	Predictive latency compensation — the first appearance of the 'video' modality
Planning × structured data	Astrobee
— (architecture not disclosed)	Dragon
— (no trained model at all)	Telesurgery over Starlink (mention)

Conclusion of Section 2

DRAGON

partially

via the exception 'enabled what was impossible without it'

PREDICTIVE COMPENSATION

column b

the effect is one publication, not statistics

ASTROBEE

column b

without caveats on the first two conditions

STARLINK TELESURGERY

does not pass

there is no trained model at all — a breakdown of a false match

Where delay and speed do not allow it — control is handed to automation in advance with the human's right to intervene, **not replaced by it entirely and not taken away from the human after the fact of an accident.**

Bridge to Section 1: at a 22-minute delay at the far end of the range, no human on Earth can be "in the loop" in the sense the Dragon commander was at a distance of 200 meters.

3. Design and new capabilities

Not autonomy of action — thinking at the drafting table, which did not exist before at all.

In order:

Surrogate models — in Raptor engine design

Claude/Anthropic — rover route planning

AIFS and AIGEFS — AI weather forecast models

Generative design — NASA Goddard structures



Stage: AI-assisted design and forecasting

TERM

AI-assisted design and forecasting

tasks that make possible what did not exist at all

DATA (4 AREAS)

engine configurations, imagery + terrain, atmosphere, geometry space

thousands of variants / route / forecast / structure

WITHOUT A MODEL

manual calculation, hours per variant

CFD (computational fluid dynamics — computing flow and combustion) for one configuration, manual planning, a supercomputer, CAD iteration by iteration

Why this section comes third, not first. Design seems like the most "obvious" niche for AI. Putting it first would hide what the section actually shows: even where distance is not a constraint, the model still does not fully replace the classical check.

Where the model gives a new capability, where its role is invisible

The model's scale of impact grows by orders of magnitude between breakdowns — from one engineering department to several real NASA mission concepts. The engineering principle does not change — **except for one explicit exception.**

In the first three breakdowns, the model predicts a number or a plan over an already-given structure. In generative design, it **produces the geometry itself**, which no one specified in advance.

How the methods are built: the surrogate model and generative search

COST OF THE SURROGATE'S LIMITATION

guaranteed only within the trained range

outside it — the prediction may be inaccurate, and there's no way to know without a full check

FROM SURROGATE TO GENERATIVE DESIGN

evaluation → production

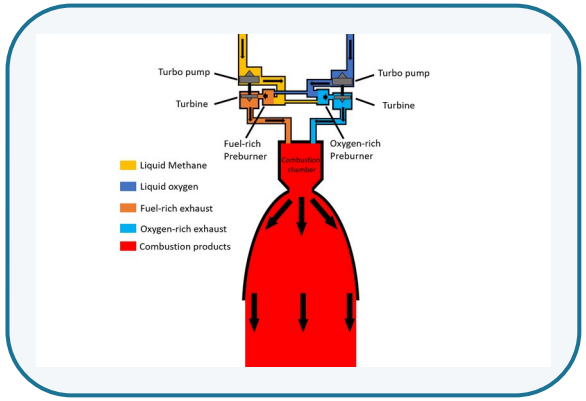
a surrogate answers 'how good is this configuration'; generative design asks 'what geometry can even exist here'

Steps

CFD simulations (expensive, hours) → Pairs of "parameters → result" as training examples → The surrogate predicts in seconds

Surrogate model: the expensive stage happens once, after that — almost free

The expensive stage does not disappear — it moves to a different point in time. It still takes computing enough examples in advance, once, for the network to learn the pattern — and those examples are exactly the same "hours of computation per variant."



How the methods are built: attention on the weather grid

LECTURE 2: ATTENTION OVER TOKENS

Q/K/V, neighbor = a position in text

'a flashlight in a dark room' — the beam is pointed at the relevant tokens

AIFS/AIGEFS: ATTENTION OVER THE ATMOSPHERE'S GRID

Q/K/V, neighbor = physical adjacency of points

a point's query is compared against the keys of physically close points

input	work	output
Query of a point 'what information do I need from my neighbors' computed for every grid point	**Keys of the neighbors** compared against the query — which neighbors are relevant neighbor = physical adjacency on the grid, not a position in text	**Values of the neighbors** the physical quantities of neighboring points a weighted sum by the match between the query and the keys

Why the architectures are 16-layer — not a coincidental number, but a trade-off between how far one point's influence needs to spread in a single pass and how much that costs computationally.

Example: surrogate models in Raptor engine design

COST OF A FULL CFD SIMULATION

hours of computation per variant

limits the search to dozens of variants per cycle

COST OF THE SURROGATE

accurate only within the trained range

outside it — the prediction is inaccurate, and there's no way to find out without a full check

CLASSICAL MODE

full CFD computation, dozens of variants per cycle

a counted baseline

MODEL (SPACEX SURROGATE)

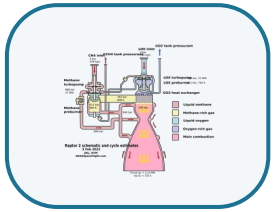
thousands of variants in the same cycle

prediction in seconds, calibrated against a subset of real CFD cases

What moved over. The engineer now sorts thousands of variants, not dozens — the stand still decides. The model widens the search, not the check.

Honest gap. SpaceX names no exact numbers for the surrogate-to-stand funnel.

Forecasting on structured data.



Job postings and publications of SpaceX's "AI for Vehicle Engineering" team

Raptor's test fire

This is exactly the final check named on the neighboring slide. The surrogate filters thousands of variants in seconds — but only a selected candidate that has passed the full CFD simulation ever reaches this kind of fire.

The fire is not an illustration of power, but a measuring instrument. This is exactly where the surrogate model's accuracy is calibrated against real combustion physics.



Wikimedia Commons, photo by Steve Jurvetson, 05/05/2020, CC BY 2.0 · Photo: Steve Jurvetson, Wikimedia Commons, CC BY 2.0 — Raptor static fire test on Starship SN4, Boca Chica

How the surrogate model is built into the Raptor engine

input

Input: Raptor configuration parameters

combustion chamber geometry, fuel-feed parameters

the range of accuracy is set by the CFD training set

work

Work: the surrogate predicts in seconds

trained on SpaceX engineers' own combustion CFD computations

predicts characteristics instead of hours of computation

output · bottleneck

Output: selection moves on to CFD check and the stand

selected configurations go through full computation and fire

accuracy is not guaranteed outside the trained range

The surrogate's design is on the neighboring slide; here — exactly what it integrates into. This same fast-evaluation technique works here for a specific Raptor engine, not an abstract example.

The network does not decide the final outcome. Selected configurations go through a full CFD computation and a static fire test — the same thing visible in the neighboring photo.

Example: Claude/Anthropic plans a rover route

COST OF MANUAL PLANNING

hours of work, proportional to route length

segment by segment from HiRISE imagery

COST OF INSUFFICIENT CHECKING

an unchecked dangerous stretch

risk of sending the rover onto unverified terrain

CLASSICAL MODE

'roughly twice as fast' — engineers' own estimate

no exact hour count for a route of the same length is in the sources

MODEL (CLAUDE, GENERAL-PURPOSE)

456 meters over 2 sols

sum of two drives (210 m + 246 m; one source rounds to ~400 m) in RML format — the same format humans use

The final check is a separate, non-AI step. JPL's digital twin ran the plan through more than **500,000 variables** for compatibility with the flight software.

The most recent breakdown in the lecture — December 2025.

Planning on multimodal data.



Anthropic + NASA/JPL, Jezero crater, December 8 and 10, 2025

The Sol 1707 image

Not an illustration, but the task's own material. It was from frames like this one from the front camera that Claude stitched together a route out of 10-meter segments and placed waypoints.

The date is not an arbitrary number. Sol 1707, December 8, 2025 — the day of the first of two drives that together covered 456 meters.



Anthropic, Claude on Mars · *Photo: Anthropic / NASA JPL — front camera of the Perseverance rover, sol 1707, December 8, 2025*

How Claude is built into route planning

input

Input: HiRISE imagery and terrain models

orbital imagery and digital terrain models of Jezero crater

work

Work: Claude plans and checks

stitches a route out of ~10 m segments, places waypoints

checks the plan for safety itself, converts it to RML format

output · threshold

Output: the digital twin, then the rover

the final check is not Claude's: >500,000 compatibility variables

classical verification, not re-modeling the terrain

The threshold sits where it doesn't appear to. Claude checks its own plan for safety — but that is not the final check; the final one belongs to the digital twin, a separate non-AI step.

RML is not a demo simplification. It's the same command format real human operators use — the plan plugs into the existing process with no rework needed.

Example: the AIFS and AIGEFS AI weather-forecast models

CLASSICAL MODE (IFS / GEFSV12)

hours on a supercomputer

more than a thousand processors, a 10-16 day forecast

MODEL (AIFS, ECMWF)

~2.5 minutes on one A100

AIFS-SUBS (46 days) — ~920x faster, ~200x less energy

MODEL (AIGEFS, NOAA, BASED ON GRAPHCAST)

<4 minutes on one A100

~0.3% of operational GFS compute cost

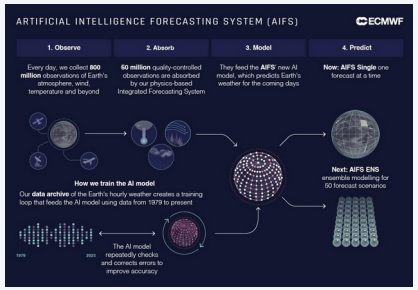
HONEST BOUNDARY

they do not do their own data assimilation

without classical 4D-Var/GDAS, which folds observations into a picture of the atmosphere, the models simply have no input

What moved over. NOAA doesn't choose between the models — it combines them: 31 runs of AIGEFS + 31 of GEFSv12 = the hybrid **HGEFS** ensemble, 62 members.

An honest ceiling nearby. Pangu-Weather is good "on average," weaker at the extremes — exactly where a forecast matters most.



[ECMWF (arXiv 2406.01465, GMD Copernicus 2026)] · [NOAA (NWS SCN 25-89)]

Status: AIFS operational

Not a photo of an office, but an official status. "AIFS operational" is an ECMWF announcement: the model has left the research stage and now counts as a forecast on par with the classical approach.

The grid in the background is not decoration. The pattern repeats the same thing: points on the atmosphere's grid, linked to each other — the design broken down on the neighboring attention slide.



ECMWF · Graphic: ECMWF — official announcement of AIFS operational status

Where AIFS and AIGEFS stand in the forecast pipeline

input · not AI

Input: classical data assimilation

4D-Var (ECMWF) or GDAS (NOAA) fold observations into an atmospheric analysis

without this step the model simply has no input

work

Work: transformer / graph network

AIFS — a 16-layer transformer with an attention mechanism

AIGEFS — a message-passing graph network on the GraphCast architecture

output

Output: a weather forecast to weather services

AIFS — ~2.5 minutes, AIGEFS — under 4 minutes, on one A100

NOAA combines AIGEFS with the classical model into an ensemble, it does not replace it

This is a different question than on the attention slide. There — how the attention mechanism works inside the model. Here — where the model itself stands among the steps AI does not perform at all.

The honest boundary is at the input, not the output. The model does not assemble the picture of the atmosphere itself: without classical data assimilation, there simply is no input for it to use.

Example: NASA Goddard generative design of structures

COST OF MANUAL CAD DESIGN

iteration after iteration by hand

four goals at once (mass, stiffness, heat, manufacturability) — hard for intuition

COST OF UNFAMILIAR GEOMETRY

harder to check before flight

an organic shape needs more engineering time to analyze

CLASSICAL MODE

manual CAD, widely practiced

no independently published number of its own — hours or cost per part

MODEL (GENERATIVE DESIGN, NASA GODDARD)

a claimed 10x reduction

the method author's own internal estimate, NOT an independently confirmed fact — applied on the Habitable Worlds Observatory

The section's fourth breakdown — the exception to the 'up to three' rule. A qualitatively new contribution: the model does not predict a number for an already-given configuration — **it produces the geometry itself.**

Generation on structured data — the first time the 'generation × structured data' cell in the lecture's grid is filled.



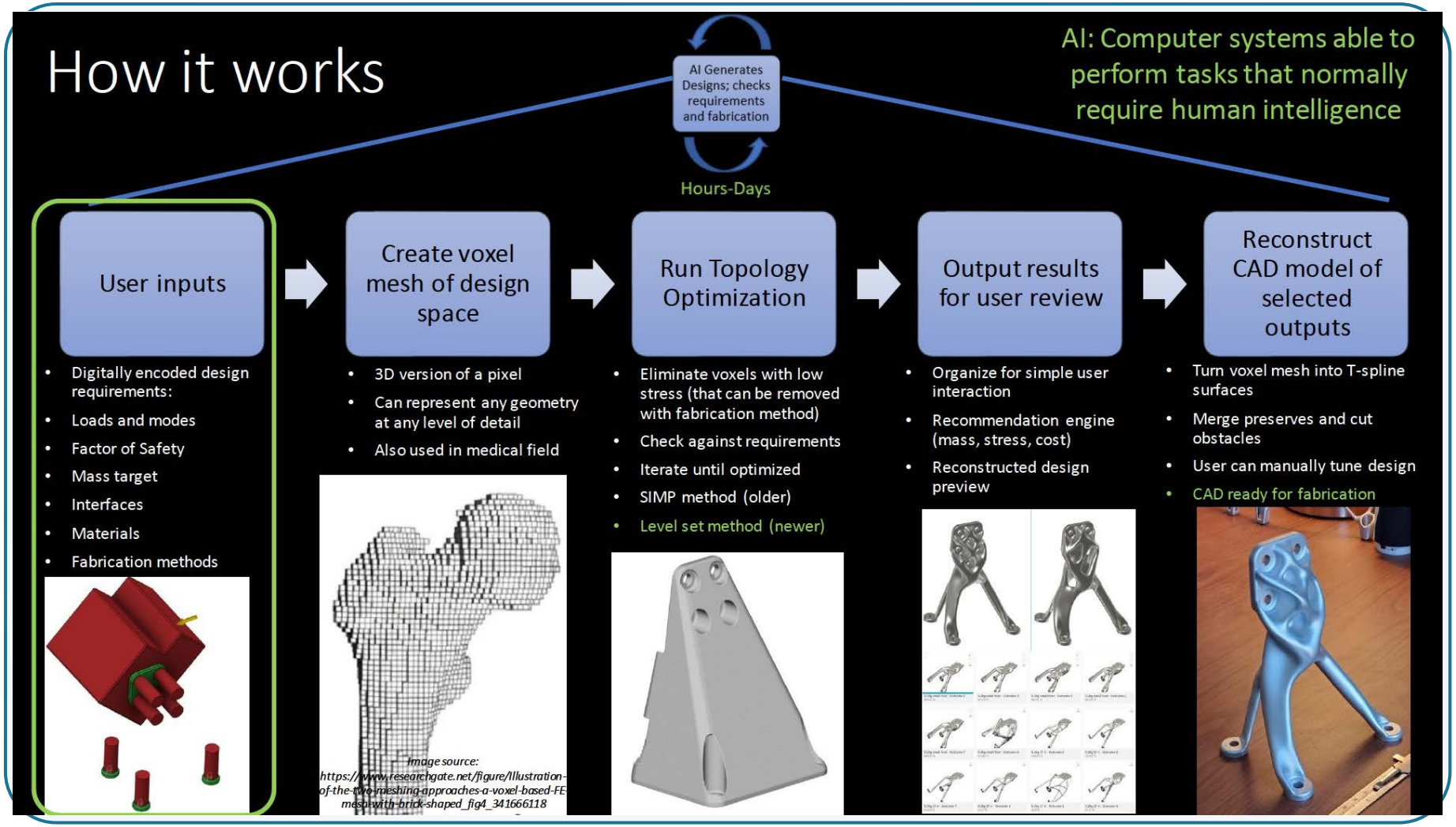
NASA NTRS (citations/20230013955, Ryan McClelland) · *Fast Company*

The five steps of the search

Five steps, not one leap from task to answer. From loads and constraints at the input to a finished shape at the output.

The engineer sets the goal, not the lines. The algorithm finds the shape within the given boundaries.

1. Loads and constraints → 2. Spatial mesh → 3. Topology optimization → 4. Checking the result → 5. CAD model

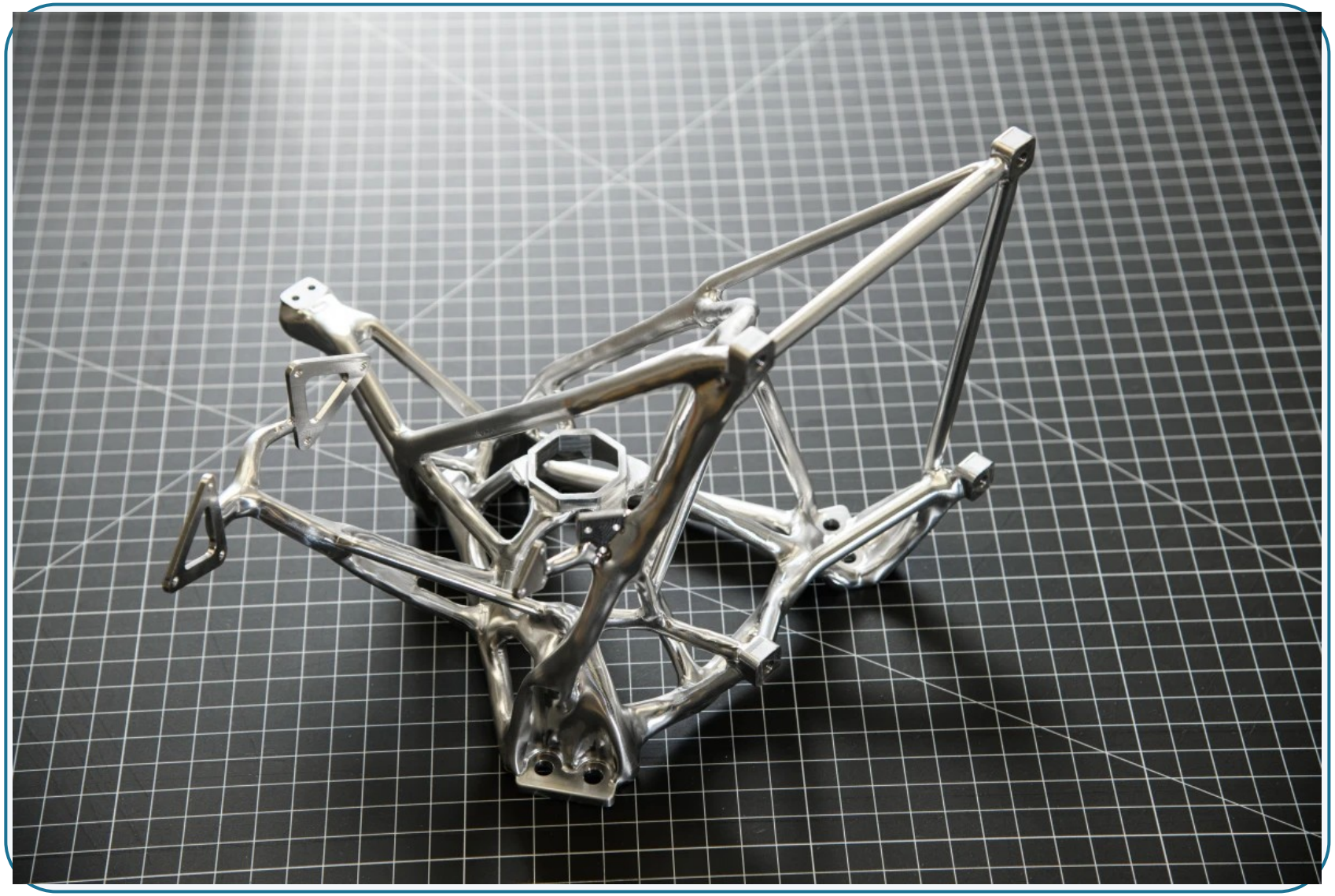


NASA NTRS (citations/20230013955, Ryan McClelland) · Slide 10 of the NASA Goddard presentation: input mesh → topology optimization → finished geometry

The organic part

A close-up of the very geometry the algorithm found. Branching, uneven ribs instead of a flat panel and a straight stiffening edge.

Not one straight edge where an engineer would normally draw a straight line. Material is removed exactly where it's not needed for the load — not wherever is convenient for a cutting tool or mill.



NASA NTRS (citations/20230013955, Ryan McClelland) · Close-up of the geometry found by the algorithm, NASA Goddard

Mentions from Section 3

- **NVIDIA FourCastNet3** — commercial wind-energy forecasting: Eni, Southwest Power Pool, Hitachi
- **Satellite + AI in agricultural insurance** — speeds up crop-loss assessment **by up to 50%**
- **Planetary defense — Sentry-II and ATLAS, already-operating systems.** Sentry-II assesses impact risk **100 years ahead**; ATLAS reduces the false-positive rate. **Honest calendar caveat:** NEO Surveyor is **delayed**, launch no earlier than September 2027 — not to be confused with the already-operating Sentry-II/ATLAS
- Short mentions (1-2 lines, the topic of other lectures in the course): satellite imagery analysis of Earth · ML exoplanet search (TESS/RAVEN) · onboard anomaly diagnostics (Artemis II, OPS-SAT)

NVIDIA · EarthDaily Agriculture · NASA (Sentry-II, ATLAS) · science.nasa.gov (NEO Surveyor)

What it gave and what it required — Section 3

Gave: orders of magnitude more variants, routes, forecasts, or geometries in the same engineer-time.

Required: training data from already-computed classical cases — the surrogate does not exist without them — and a separate, non-AI checking step before trusting the result.

Breakdown	Risk of a model error
Raptor surrogate	a few hours lost — filtered out by the stand
Claude/rover route	a lost planning cycle — filtered out by the digital twin
AIFS/AIGEFS	the highest cost — a real decision affecting many people
Generative design	intermediate — an unfamiliar shape is costlier to check

Conclusion of Section 3

RAPTOR SURROGATE

column a

all three conditions, but without SpaceX's own numbers

CLAUDE/ROUTE

column b

second unconditional, a caveat on the third

AIFS/AIGEFS

column a

the most solid confirmation in the section

GENERATIVE DESIGN

column a

the first condition in its strongest form

A general- or special-purpose model expands how many options can be considered per unit of time — or, for generative design, expands the space of possible options itself. **It does not replace the final check in any of the four cases.**

Bridge to Section 4: if a model can work from anywhere, with no physical tie to a location — where is it physically most advantageous to place it?

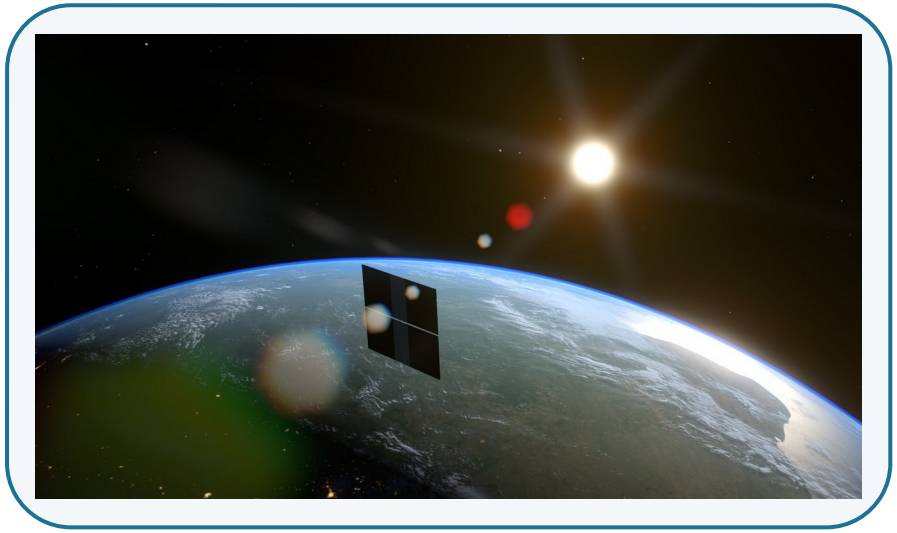
4. Space for AI

A mirror of the whole lecture: it isn't that space is making AI smarter — AI is physically moving into space.

In order:

Starcloud — the first data-center-class GPU in space

Google Project Suncatcher — a satellite with tensor processing units



Stage: orbital computing

TERM

orbital computing

placing AI computing infrastructure physically in space, not on Earth

DATA

a data center's power and cooling budget

not observation of the outside world — an internal budget of the infrastructure itself

WITHOUT ORBITAL PLACEMENT

a ground-based data center

limited by the power grid's throughput and the region's climate

A reversal of the whole lecture's logic. In Sections 1–3, the question was "what part of a human's work can a model do next to the action itself." Here the question is different: not what the model can do, but **where it's physically cheaper to put it.**

Why it's inference, not training, that moves to space

Training — why it stays on Earth

- Thousands of accelerators compute one model and synchronize at every step
- Every step changes the model's weights in their entirety, and the copies cannot be allowed to diverge
- Power and networking are sized for the whole cluster at once, not for a single chip

Inference — why we move it to orbit

- One pass of the finished network from input to answer, nobody touches the weights
- A neighboring accelerator for step-by-step synchronization isn't needed at all
- Starcloud-1 — a single H100 chip answers requests to the Gemma model

The difference isn't in what's being computed, but in how many accelerators have to talk to each other at every step. Training needs thousands; inference needs one or a few, with no constant cross-checking between them.

On Starcloud-1, the Gemma model's weights don't change by a single step. The chip runs a finished network — it doesn't train a new one.

What already works, and what is still demonstrator-sized

This is the start of a new infrastructure category, not a replacement for the existing one — and the breakdowns that follow say exactly that, without overstating it.

In space right now: **5 chips** — *1×H100 (Starcloud-1) + 4×TPU (Suncatcher) — s52/s53*



In a single ground-based AI data center: **tens of thousands of GPUs** — *typical size of a modern training cluster*



The bars are not to linear scale — the difference is orders of magnitude, not percentage points.

Example: Starcloud — the first data-center-class GPU in space

COST OF A GROUND-BASED DATA CENTER

cheap to build

limited by the power grid's throughput and climate

COST OF AN ORBITAL DATA CENTER

launch and upkeep cost orders of magnitude more

near-continuous sunlight is the main resource gained in return

CLASSICAL MODE

does not exist

'a data center in space' is a new category, not a faster version of the old one

MODEL/AUTOMATION (STARCLOUD-1, 60 KG)

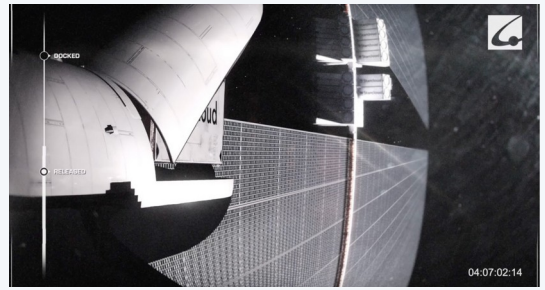
launched 02.11.2025

first data-center-class H100 in space; serving requests to the Gemma model

An FCC (the US telecom regulator) filing for 88,000 satellites is the company's plan, not a fact. Series A round of $170 million (03.2026) + $250 million at a $2.3 billion valuation (08.2026).

What moved over. The question "where is land and electricity cheaper" moved into the question "where is the sun more continuous."

Text generation.



CNBC, 02.11.2025

The Starcloud team

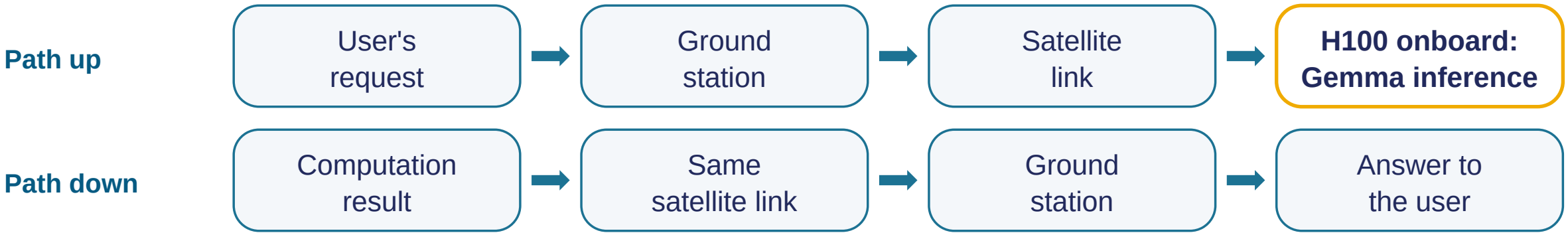
Real people stand behind the satellite, not just an engineering render. This exact team took the demonstrator from idea to a working H100 in orbit.

NVIDIA's Inception program isn't a marketing detail. As a participant in this startup program, Starcloud gets access to NVIDIA's hardware and expertise.



NVIDIA blog, Starcloud · *Photo: NVIDIA — the Starcloud team*

How a request's path through Starcloud is built



The request's path up and the answer's path down — the computation inside the chip doesn't differ from Earth's

What's new is only at the input and output. The satellite link up and down — that's the entire difference from an ordinary ground-based data center.

Inside the chip — nothing new. Inference on the Gemma model aboard the H100 runs exactly as it would on Earth: the difference is in the location and the power source, not in the computation.

Example: Google Project Suncatcher

COST

the same logic as Starcloud

a different company, a different chip

CLASSICAL MODE

does not exist

the same gap, for the same reason

MODEL/AUTOMATION (SATELLITE + 4 TPUS)

launched 01.10.2026

first AI model response from a data-center-class TPU in orbit

COMPANY'S PLAN

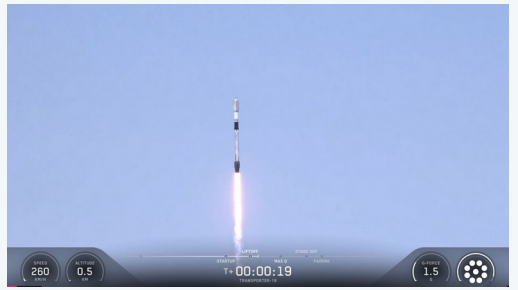
81 satellites in a cluster ~1 km in radius

laser inter-satellite links; partnership with Planet Labs — 2 satellites by early 2027

The engineering sense of the name: "sun catcher" — not cooling, not radiation shielding, but the continuity of sunlight as the main resource.

The company openly admits: some engineering decisions can only be verified in real orbital conditions, not on Earth.

General-purpose generation/inference — an infrastructure demonstrator, not tied to a single task.

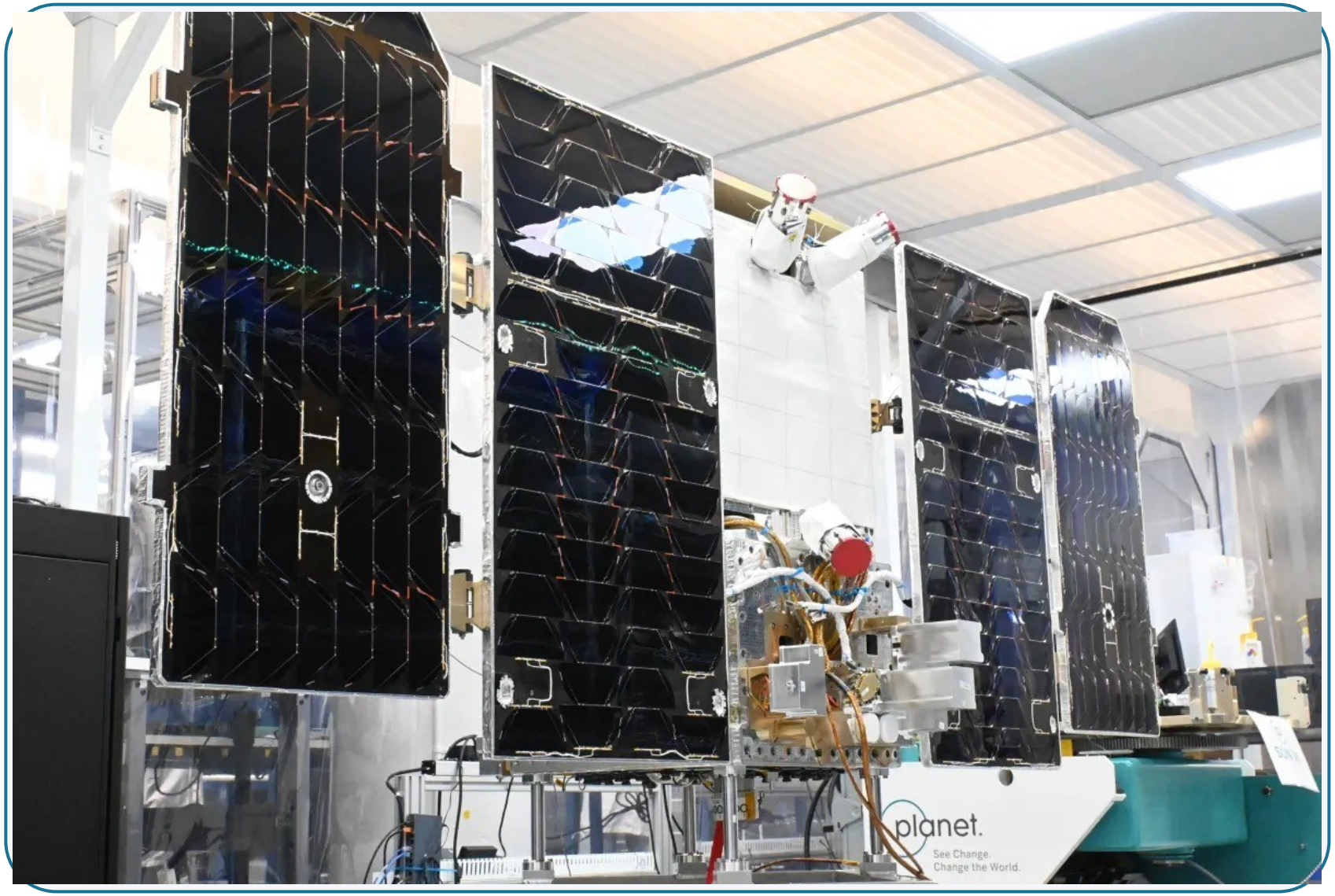


[CNN, 01.10.2026]

Suncatcher during assembly

Real hardware, not concept art. The deployed solar panels and the satellite body are what physically flew, assembled in the assembly facility before launch.

Inside are four tensor processing units. The same physical unit that has already delivered the first AI model response in orbit.



TechCrunch, 30.10.2026 · Photo: the assembled Suncatcher satellite in the assembly facility

Suncatcher's second AI case: the model holds formation

input · not AI

Input: the formation's orbital dynamics

position and velocity of every satellite in the formation, in real time

numerical integration of the equations of motion — classical mechanics

work

Work: differentiable optimization of the formation

objective function — the formation's geometry: a cluster ~1 km across with no collisions

gradient — adjoint-state methods

reverse-mode automatic differentiation (AD) in JAX — into the model

output

Output: a maneuver command for every satellite

a corrected maneuver command for every satellite in the formation

keeps the cluster at a radius of about 1 km with no collisions

Suncatcher's second AI case isn't inference — it's formation control. The tighter the formation, the higher the bandwidth of the laser links between satellites.

The mechanism is differentiable optimization, not prediction from examples. The gradient runs backward through the motion's integration — into the model's parameters, the same way neural networks are trained.

Google, Project Suncatcher preprint — Supplementary Information, Orbital dynamics modeling

Mentions from Section 4

- **Starcloud's FCC filing for 88,000 satellites** — the company's plan, **not a fact of today**
- **A reminder from Section 0:** the ~12x growth in satellites over 10 years and the multi-fold growth in data are the direct source of the computing demand this section describes

What it gave and what it required — Section 4

Gave: proof that AI hardware physically works and answers real requests in space.

Required: solving problems that don't exist on Earth — radiation hardening, heat rejection in vacuum with no convection, servicing with no human access — and a willingness to test some of the solutions only in actual flight.

Section	Cells occupied (cumulative)
After Section 1	3
After Section 2	5
After Section 3	8
After Section 4	9 (of the 6 task types, 5 are occupied — retrieval never came up)

Conclusion of Section 4

SECTION 1

machine → space

to act where a human cannot be

SECTION 4

computation → space

to be where there is more energy

A mirror process relative to Section 1. There, the machine moved into space to act where a human cannot be. Here, computation moves into space to be where there's more energy.

Next — Assembly brings all four fronts of the lecture together into a single answer.

Section 5. Assembly

The four substantive sections are done. Assembly brings the material already shown together in three different ways:

- **Registry** — the denominator and the exceptions
- **Answer to the lecture's question** — section by section, by name
- **Final summary** — the count of thirteen breakdowns, the honest gaps, and where AI is flatly not needed

13 + 18 = 31

Registry of breakdowns: the denominator and the exceptions

13 detailed breakdowns + 18 short mentions = 31 AI-in-space cases. This is the stated denominator: before trusting any percentage or conclusion later in the lecture, it's clear how many cases it's based on.

Checkability: each of the 13 breakdowns names a source on its slide — it can be opened and checked against what the lecture says.

Section	Detailed breakdowns	Short mentions	Distinguishing feature
1	4	6	LeoLabs — a separate breakdown: answers the question left open at the start of the lecture
2	3	4	—
3	4	6	Generative design — a qualitatively new type of model contribution
4	2	2	—

Answer to the lecture's question — Sections 1 and 2

Section 1. Geometric navigation — carried over ready-made. The SLAM architecture for subsurface exploration (NeBula/LAMP) — **wasn't born in space**: invented for an Earth-based mine, then adapted for a lunar tube.

Section 2. Teleoperation as a principle — carried over ready-made. Telesurgery over Starlink — space created a **new engineering configuration** that didn't exist before: not through a model, but through communication infrastructure.

2 : 2

Answer to the lecture's question — Sections 3 and 4

Section 3. Surrogate models and generative search — general ML techniques, not an invention of space. But the specific application (Claude on Mars, geometry for the flagship telescope) — was physically impossible before powerful general-purpose models appeared.

Section 4. The idea of putting computing next to the power source isn't new. But the engineering of an orbital data center — **didn't exist before this specific project.**

Overall count: 2:2 — not a target set in advance, but an honest outcome of selecting the material.

13 of 13

The count: thirteen breakdowns, one boundary, never crossed

FINAL CHECK OR THE RIGHT TO INTERVENE

rests with a classical, non-AI step

in all thirteen cases without exception

MODEL DOES SUBSTANTIVE WORK

in 12 of 13

the exception is Dragon, whose architecture isn't publicly disclosed

AEGIS — a target profile set in advance · **SPOC** — the human driver, the final authority · **COROB-X** — the EaRLaP pipeline + after-the-fact review · **LeoLabs** — the operator decides the maneuver, not the model · **Dragon** — manual intervention · **Predictive latency compensation** — the operator decides for himself · **Astrobee** — a classical controller refines it · **Raptor surrogate** — the test stand and CFD · **Claude/route planning** — a digital twin, 500,000 variables · **AIFS/AIGEFS** — classical data assimilation · **Generative design** — engineering qualification before flight · **Starcloud/Suncatcher** — the decisions made by people at the design stage.

Summary of honest gaps

- **Operating budget and decision-revision protocol** — the number of ChemCam measurements per cycle, IPEx's behavior in an off-nominal situation, manual intervention in Dragon at the last moment
- **Feedback loop between outcomes and system updates** — whether AEGIS's profile gets revised, whether COROB-X's quality cycle closes
- **Composition and volume of training data** — not disclosed for AEGIS, COROB-X's CNN, Astrobee, LeoLabs; contrast with AIFS/AIGEFS and SPOC, which disclose it in detail
- **System architecture, not disclosed by the vendor** — the NASA Docking System (Dragon), ICON's quality-control layer
- **Economics at the level of a single operation** — Starcloud/Suncatcher don't publish per-request cost; generative design's 10x is an internal estimate
- **The step after a warning, outside the organization's control** — LeoLabs has no data on operators' actual maneuvers

Where AI is flatly not needed

Point 5 — honestly not found in this lecture's material. That doesn't mean such a case doesn't exist in the industry at all — it means it didn't come up within the thirteen cases selected.

Criterion	Example from the lecture
The task is fully described by known physics	AutoNav/GESTALT, G-FOLD, Terrain-Relative Navigation, NASA IPEx
The decisive leap is in the infrastructure, not the algorithm	telesurgery over Starlink
The task is new — classical methods simply have nothing to build an alternative from	COROB-X, telesurgery, orbital data centers
The final check stays classical regardless of the model	all of Section 3 is 'not enough alone', not 'not needed'
The model creates dependency with no proportionate payoff	didn't come up in the 13 breakdowns — named honestly

A debt carried forward: certifying autonomous systems

BREAKDOWNS UNDER FORMAL CERTIFICATION OF AUTONOMOUS BEHAVIOR SPECIFICALLY

0 of 13

the component gets certified, the campaign gets tested — not the autonomous decision itself

Every organization has its own strict internal approval procedure. But a common, cross-agency, international standard specifically for autonomous behavior — unlike the certification standards for the aircraft or spacecraft itself — **doesn't exist.**

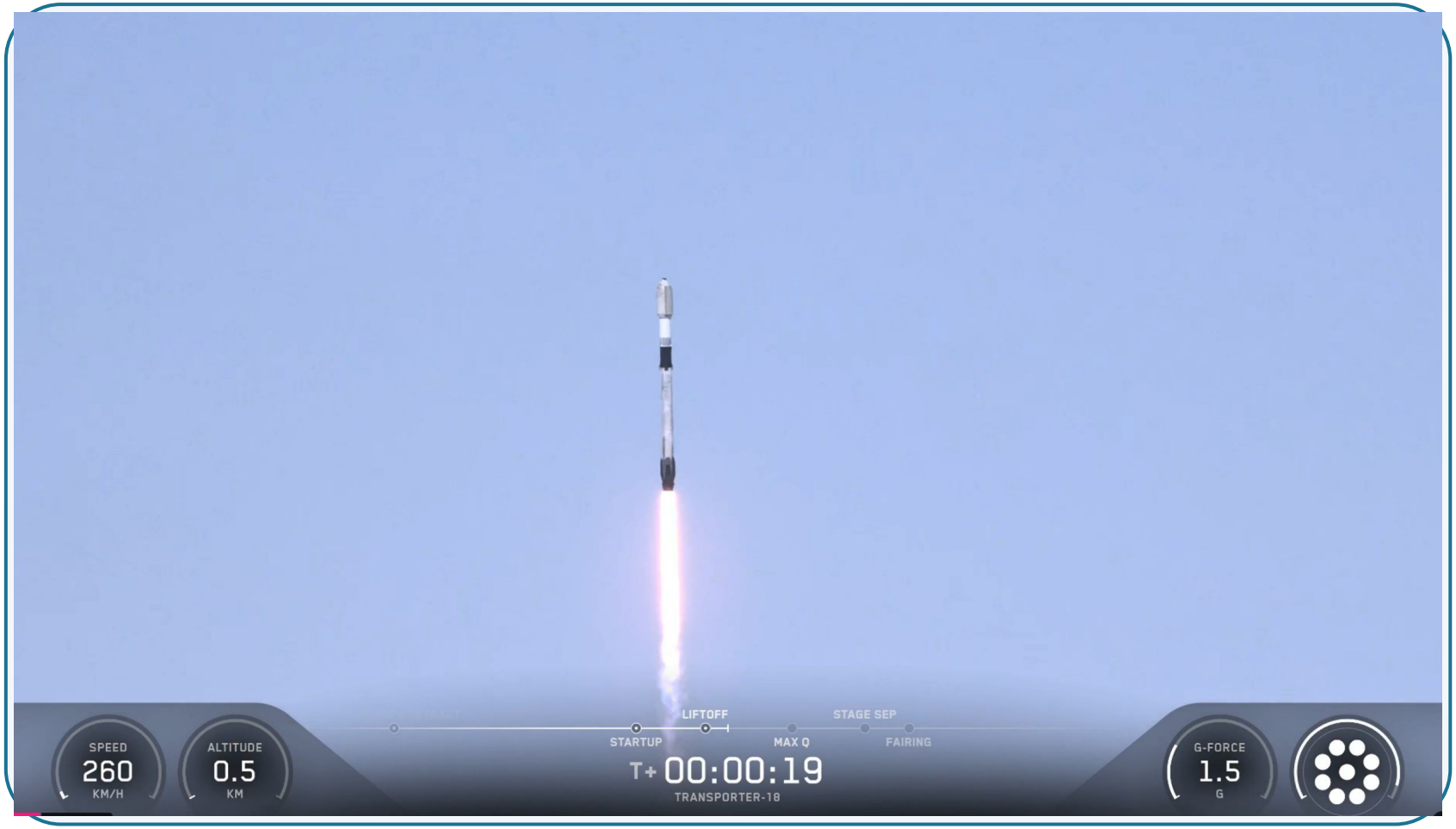
This isn't specific to space — it's a general feature of every technology that develops faster than the institutional structure meant to regulate it.

An earned claim

The familiar answer: solutions come from the lab, space just applies them.

The earned claim: the frontier itself forced the invention of something that didn't exist, and showed that it works.

The course moves on, to agriculture: who to trust with the decision — the model, classical methods, or the human who signs off on the result.



Questions